

VirtualBox

Category: Virtualization • Cybersecurity Lab

Purpose: Learn how VirtualBox supports isolated cybersecurity labs, how virtual networking works, and how to safely organize Kali Linux, Windows, and vulnerable test machines.

1. What is VirtualBox?

VirtualBox is a general-purpose virtualization platform that allows operating systems to run as virtual machines (VMs) on a host computer. A VM provides virtual hardware such as CPU, memory, storage, network adapters, and display devices.

2. Host vs Guest

Term	Meaning
Host OS	The operating system installed directly on the physical computer.
Guest OS	The operating system running inside a virtual machine.
VM	A software-defined computer with virtualized hardware resources.
Hypervisor	Software layer that manages virtual machines and their access to physical resources.

3. Why VirtualBox is Useful for Cybersecurity

- Create isolated practice environments.
- Run multiple operating systems on one physical machine.
- Build repeatable labs for networking, reconnaissance, exploitation, and defense.
- Take snapshots before risky lab changes and restore the VM afterward.
- Connect VMs through controlled virtual networks.

4. Common VM Components

Component	Purpose
CPU	Provides virtual processor resources.
RAM	Provides memory to the guest OS.
Virtual disk	Stores the guest operating system and files.
Network adapter	Connects the VM to a virtual or physical network.
ISO image	Installation media used to install an operating system.
Snapshot	Stores a restore point for a VM's state and disk configuration.

5. VirtualBox Network Modes

Mode	Typical use
NAT	VM accesses external networks through the host's network connection. Useful for ordinary internet access.
Bridged Adapter	VM appears as a device on the same physical network as the host.
Host-only Adapter	Creates communication between the host and VMs without requiring external network access.
Internal Network	Allows VMs attached to the same named internal network to communicate with each other.

6. Cybersecurity Lab Network

For an isolated practice lab, Internal Network or Host-only networking can be used depending on whether the host must communicate with the VMs. For example, Kali Linux, Windows 7, and Metasploitable can be placed on the same isolated virtual network for controlled testing.

Keep intentionally vulnerable VMs separated from your normal home or office network. Avoid attaching vulnerable machines directly to a production network.

7. Example Lab Architecture

VM	Role
Kali Linux	Security testing and analysis machine.
Windows test VM	Client/endpoint for security exercises.
Metasploitable	Intentionally vulnerable target for controlled practice.
Virtual network	Isolated communication path between the lab machines.

8. VM Installation Workflow

- Download a legitimate OS ISO from its official source.
- Create a new VM and select the appropriate guest OS type.
- Assign reasonable CPU and RAM without starving the host.
- Create or attach a virtual disk.
- Attach the ISO and complete OS installation.
- Install Guest Additions when appropriate and supported.
- Configure the VM's network adapter according to the lab design.
- Take a clean snapshot before major experiments.

9. Snapshots

A snapshot records a point in the VM's state so you can return to it later. Snapshots are useful before security exercises that may change system configuration. They are not a replacement for independent backups.

10. VirtualBox Security Considerations

- Keep vulnerable VMs on isolated networks.
- Do not expose intentionally vulnerable services to the public internet.
- Use snapshots before major lab changes.
- Avoid sharing sensitive host files with untrusted guest systems.
- Review shared clipboard and drag-and-drop settings when isolation matters.
- Use strong host-account security because compromise of the host can affect the lab.

11. Useful Checks Inside Linux VMs

```
ip addr
```

Display network interfaces and IP addresses.

```
ip route
```

Display the guest's routing table.

```
ping <IP>
```

Test connectivity to another authorized lab machine.

```
ip neigh
```

View the local neighbor table.

12. Troubleshooting Lab Connectivity

- Confirm the VMs use the intended network mode.
- Check that both VMs are attached to the same Internal Network name when using Internal Network.
- Check IP addresses and subnet configuration.
- Verify that the guest firewall is not blocking the test traffic.
- Check the routing table when communication crosses different networks.
- Avoid assuming that identical-looking adapters automatically provide the same network connectivity.

13. Common Mistakes

- Giving a vulnerable VM direct access to the internet unnecessarily.
- Mixing isolated lab adapters with production networking without understanding the routing.
- Assigning insufficient RAM or CPU to the guest.
- Forgetting which virtual adapter is connected to which network.
- Relying on snapshots as the only form of backup.
- Changing several network settings at once, making troubleshooting difficult.

14. Quick Revision

Concept	Remember
Host	Physical computer running VirtualBox.
Guest	Operating system inside a VM.
NAT	VM uses the host's network connection for external access.
Host-only	Host and selected VMs can communicate on a private virtual network.
Internal Network	VM-to-VM communication on a named isolated virtual network.
Snapshot	Restore point for a VM state.

15. Key Takeaways

- VirtualBox is useful for building controlled cybersecurity labs.
- Virtual networking determines how VMs communicate with one another, the host, and external networks.
- Isolation is especially important when using intentionally vulnerable machines.
- Snapshots make it easier to recover from lab changes.